Analog IIT Kharagpur

4/5 80%

General Aptitude

General Aptitude test

Q2.1

My grandson is about as many days as my son is in weeks and my grandson

is as many months as I am in years. My grandson, my son and I together are 160 yrs.

Can you tell my age?

S2.1 Ans (yrs) [Enter Number only]

44

Q2.2

A and B starts from the same point at opposite direction. They will move 6km

and take 8km left. How far is A and B from each other?

S2.2 **Ans (km) [Enter Number only]**

100Km

Time left in this section **00 : 39 : 25**

Q2.3

The ratio between the length and breadth of a rectangular park is 3 : 2.

If a man cycling along the boundary of the park at the speed of 12 km/ h completes

one round in 8 minutes, then the area of the park in sq. m is

S2.3 **Ans [Enter Number only]**

153600 sqm

Q2.4

Two taps can separately fill a cistern in 10 minutes and 15 minutes respectively.

If these two pipes and a waste pipe are kept open simultaneously, the cistern gets

filled in 18 minutes. The waste pipe can empty the full cistern in

S2.4 **Ans (minutes) [Enter Number only]**

Q2.5

A metal sheet 27 cm long, 8 cm broad and 1 cm thick is melted into a cube.

The difference between the surface area of the two solids, is

S2.5 **Ans (cm) [Enter Number only]**

286

Q2.6

S2.6 What Comes next in the series RL12, TQ20, WU32, AX48, ?

EZ68

Q2.7

Corners are cut off from an equilateral triangle T to produce a regular hexagon H.

Then, the ratio area of H to the area of T is:

S2.7 **Ans [Enter Number only]**

Q2.8

The number of girls appearing for an admission test is twice the number of boys.

If 30% of the girls and 45% of the boys get admission, the percentage of candidates

who do not get admission is:

S2.8 **Ans (%) [Enter Number only]**

Q2.9

Angad was conducting an experiment in which the average of 11 observations

came to be 90, while the average of first five observations was 87, and that of the

last five was 84. What is the measure of the 6th observation?

S2.9 **Ans [Enter Number only]**

Q2.10

A car starts running with the initial speed of 40 kmph, with its speed increasing every hour by 5 kmph. How many hours will it take to cover a distance of 385 km?

S2.10 **Ans (hrs) [Enter Number only]**

Q2.11

S2.11 ABCD is related to OPQR in the same way as WXYZ is related to

Q2.12

S2.12 A blacksmith has five iron articles A, B, C, D and E, each having a different weight.

(i) A weighs twice as much as B.

(ii) B weighs four-and- a-half times as much as C.

(iii) C weighs half as much as D.

(iv) D weighs half as much as E.

(v) E weighs less than A but more than C.

The lightest in weight is [eg:A]

E is lighter in weight than these two articles [eg:A,B]

E is heavier in weight than these two articles

The heaviest in weight is

Q2.13

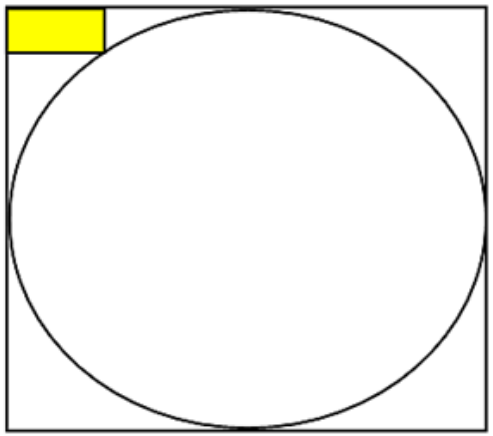
S2.13 Given a<b<c<d , what is the maximum ratio of given equation?

- (a+b)/(c+d)
- (b+c)/(a+d)
- (c+d)/(a+b)
- (a+c)/(b+d)

Q2.14

In the below figure, the inner rectangle measures 5cm x 10cm.

What is the radius of the circle in cm.



S2.14 **Ans (cm) [Enter Number only]**

Q2.15

If the difference between two numbers is 9 and their product is 19, what is

the sum of their squares.

S2.15 **Ans [Enter Number only]**

Q2.16

If 8 monkeys take 8 minutes to eat 8 bananas, How many monkeys would it

take to eat 40 bananas in 80 minutes.

S2.16 **Ans [Enter Number only]**

Q2.17

A is twice as efficient as B. Together, A and B can finish a project in 5 days.

In how many days can A working alone complete the work?

S2.17 **Ans [Enter Number only]**

Q2.18

Compute the result of $$~30 - $~5 if $ were to denote the “square of ” operator

and ~ were to denote “square root of ” operator

S2.18 **Ans [Enter Number only]**

Q2.19

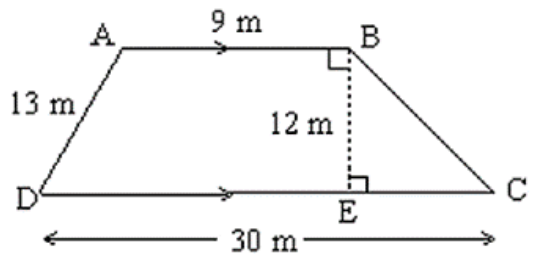
If each child is given 10 sweets, there are 3 sweets left over. But if each is given 11,

then the number of sweet is 4 less. Find the number of sweets

S2.19 **Ans: [Enter Number only]**

Q2.20

A plot of land is in the shape of a trapezium whose dimensions are given in the figure below :



Hence the perimeter of the field is _____

S2.20 **Ans (m) [Enter Number only]**

Q2.21

A man decides to buy a nice horse. He pays Rs 60 for it, and he is very content with the strong animal. After a year, the value of the horse has increased to Rs 70 and he decides to sell the horse. But already a few days later he regrets his decision to sell the beautiful horse, and he buys it again. Unfortunately he has to pay Rs 80 to get it back. After another year of owning the horse, he finally decides to sell the horse for Rs 90. What is the overall profit the man makes?

S2.21 **Ans (Rs) [Enter Number only]**

Q2.22

A bucket contains 200cc of liquid. A solid ball is dropped in the bucket resulting in the rise of liquid level to 1.3 times of its original level. If the radius of the base of the bucket is 3 cm and the radius of the surface of the liquid level is 1 cm more than the radius of the base of the bucket before the ball is dropped.

S2.22 Find the volume of the solid metal ball.

- 68cc
- 80cc
- 92cc
- Cannot be determined

Q2.23

In a survey on the viewership of the TV channels, 73% of those surveyed viewed at least one of the three Channels: Star Plus, Sab TV, and Sony. 38% of those surveyed viewed Star Plus, 39% viewed Sony, and 23% viewed Sab TV. 11% of all those surveyed viewed all the three channels.

S2.23 What percentage of those surveyed, viewed more than one of the three TV channels?

- 16
- 38
- 27
- Cannot be determined

Q2.24

Joseph is in a dilemma. He has been offered a job which would pay him Rs 80,000 per month for first three years and Rs 1,20,000 per month for the next three years, and Rs 1,50,000 per month for the remaining four years. He has also been offered an MBA at a prestigious place and he is considering whether to accept the job or go for the MBA. The first year tuition fee for the MBA program is Rs 16,00,000 and the second year tuition fee for the MBA program is Rs 20,00,000. After MBA, he'll get a salary of Rs 2,00,000 per month for the first four years and then Rs 2,50,000 per month for the remaining four years.

S2.24 What will be the approximate percentage gain for Joseph in opting for the MBA instead of the job in the 10 years horizon considering no discounting of money?

- 23
- 25
- 27
- 29

Q2.25

S2.25 The number 37^371 - 26^371 is divisible by

- 10
- 11
- 12
- 15

Q2.26

S2.26 Consider a class of 40 students whose average weight is 40 kgs. "m" new students join this class whose average weight is "n" kgs. If it is known that m + n = 50, what is the maximum possible average weight of the class now?

- 40.18
- 40.56
- 40.67
- Cannot be determined

Q2.27

Janta Airline has a free luggage allowance for its passengers. If any passenger carries excess luggage, it is charged at a constant rate per kg. The total luggage charge paid by Ravind Jekriwal and Pranas Shubhan is Rs. 1100. If both Ravind and Pranas had carried luggage twice the weight than they actually did, their luggage charges would have been Rs. 2000 and Rs. 1000 respectively.

S2.27 What was the charge levied on Ravind's luggage?

- 800
- 700
- 600
- 900

Q2.28

The average age of a couple was 24 years. After their 1st and 2nd children (twins) were born, the average age of the family became 13.5 years. The average age of the family just after 3rd child was born was 13.2 years. The average age of the family after 4th child was born was 16 years. The current average age of the family is 19 years.

S2.28 What is the current age of the twin children?

- 10 yrs
- 12 yrs
- 13 yrs
- 11 yrs

Q2.29

A fruit seller has oranges, apples and guavas in the ratio 2:5:8. The number of apples is more than

the number of oranges by a number that is a multiple of both 6 and 8.

S2.29 What is the minimum number of fruits in his shop?

- 240
- 360
- 120
- 90

Q2.30

A 20% ethanol solution is mixed with another ethanol solution, say, S of unknown concentration in the proportion 1:3 by volume. This mixture is then mixed with an equal volume of 20% ethanol solution. If the resultant mixture is a 31.25% ethanol solution, then S2.30 the unknown concentration of S is?

- 50%
- 55%
- 48%
- 52%

Q2.31

A man standing on top of a tower sees a car coming towards the tower. If it takes 20 minutes for the angle of depression to change from 30° to 60°, S2.31 what is the time remaining for the car to reach the tower?

- $20\sqrt{3}$ minutes
- 10 minutes
- $10\sqrt{3}$ minutes
- 5 minutes

Q2.32

S2.32 Read the below data and answer the 4 Questions below.

A tea taster was assigned to rate teas from six different locations - Munnar, Wayanad, Ooty, Darjeeling, Assam and Himachal. These teas were placed in six cups, numbered 1 to 6, not necessarily in the same order. The tea taster was asked to rate these teas on the strength of their flavour on a scale of 1 to 10. He gave a unique integer rating to each tea. Some other information is given below:

1. Cup 6 contained tea from Himachal.

2. Tea from Ooty got the highest rating, but it was not in Cup 3.

3. The rating of tea in Cup 3 was double the rating of the tea in Cup 5.

4. Only two cups got ratings in even numbers.

5. Cup 2 got the minimum rating and this rating was an even number.

6. Tea in Cup 3 got a higher rating than that in Cup 1.

7. The rating of tea from Wayanad was more than the rating of tea from Munnar, but less than that from Assam.

		A	B	C	D
Q1	What was the second highest rating given?	8	6	7	9
Q2	What was the number of the cup that contained tea from Ooty?	3	6	4	2
Q3	If the tea from Munnar did not get the minimum rating, what was the rating of the tea from Wayanad?	3	1	5	6
Q4	If the cup containg teas from Wayanad and Ooty had consecutive numbers, which of the following may be true ?	Cup 5 contains tea from Assam	Cup 1 contains tea from Darjeeling	Tea from Wayanad has got a rating of 6	Tea from Darjeeling got the minimum rating

	A	B	C	D
Q1	○	○	○	○
Q2	○	○	○	○
Q3	○	○	○	○
Q4	○	○	○	○

Q2.33

The sequence $a_1, a_2, a_3, \cdots, a_n, \cdots$ is such that $a_n = \frac{a_{n-1} + a_{n-2}}{2}$ for all $n \geq 3$. If $a_3 = 4$ and $a_5 = 20$ what is the value of a_6 ?

S2.33 Ans

- 12
- 16
- 20
- 24
- 28

Q2.34

The average (arithmetic mean) of the integers from 200 to 400, inclusive, is how

much greater than the average of the integers from 50 to 100, inclusive?

S2.34 **Ans**

- 150
- 175
- 200
- 225
- 300

Q2.35

A closed cylindrical tank contains 36pi (36 x 3.14) cubic feet of water and is filed to half its capacity. When the tank is placed upright on its circular base on level ground, the height of the water in the tank is 4 feet. When the tank is placed on its side on level ground, what is the height, in feet, of the surface of the water above the ground ?

S2.35 **Ans**

- 2
- 3
- 4
- 6
- 9

Q2.36

A researcher computed the mean, the median, and the standard deviation for a set of performance scores. If 5 were to be added to each score, which of these three statistics would change?

S2.36 **Ans**

- ◯ The mean only
- ◯ The median only
- ◯ The standard deviation only
- ◯ The mean and the median
- ◯ The mean and the standard deviation

Q2.37

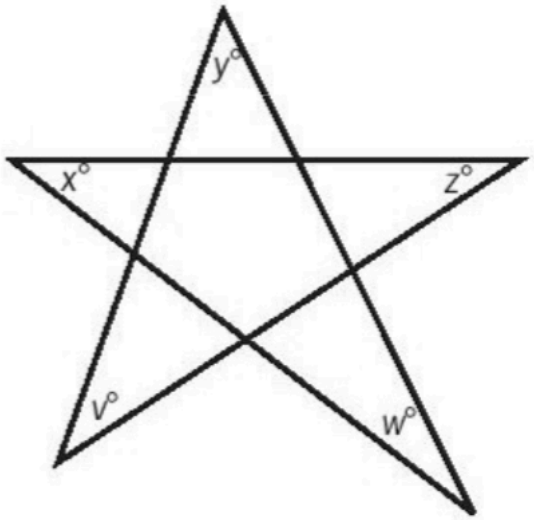
Of the three-digit integers greater than 700, how many have two digits that are qual to each other

and the remaining digit different from the other two?

S2.37 **Ans**

- ◯ 90
- ◯ 82
- ◯ 80
- ◯ 45
- ◯ 36

Q2.38



In the figure shown, what is the value of $v + x + y + z + w$?

S2.38 **Ans**

- 45
- 90
- 180
- 270
- 360

Q2.39

Positive integer y is 50 percent of 50 percent of positive integer x,

and y percent of x equals 100. What is the value of x?

S2.39 **Ans**

- 50
- 100
- 200
- 1000
- 2000

Q2.40

The product of all the prime numbers less than 20 is closest to which of the following powers of 10?

S2.40 **Ans**

- 10^9
- 10^8
- 10^7
- 10^6
- 10^5

Q2.41

If

sqrt(3 - 2x) = sqrt(2x) + 1

then 4x^2 =

S2.41 **Ans**

- 1
- 4
- 2 - 2x
- 4x - 2
- 6x - 1

Q2.42

Arun will jog from home at "x" km/hr and then walk back home by the same route at "y" km per hour.

How many km from home can Arun jog so that he spends a total of "t" hours jogging and walking?

S2.42 **Ans**

- xt/y
- (x + t)/xy
- xyt/(x + y)
- (x + y + t)/xy
- (y + t)/x - t/y

Next